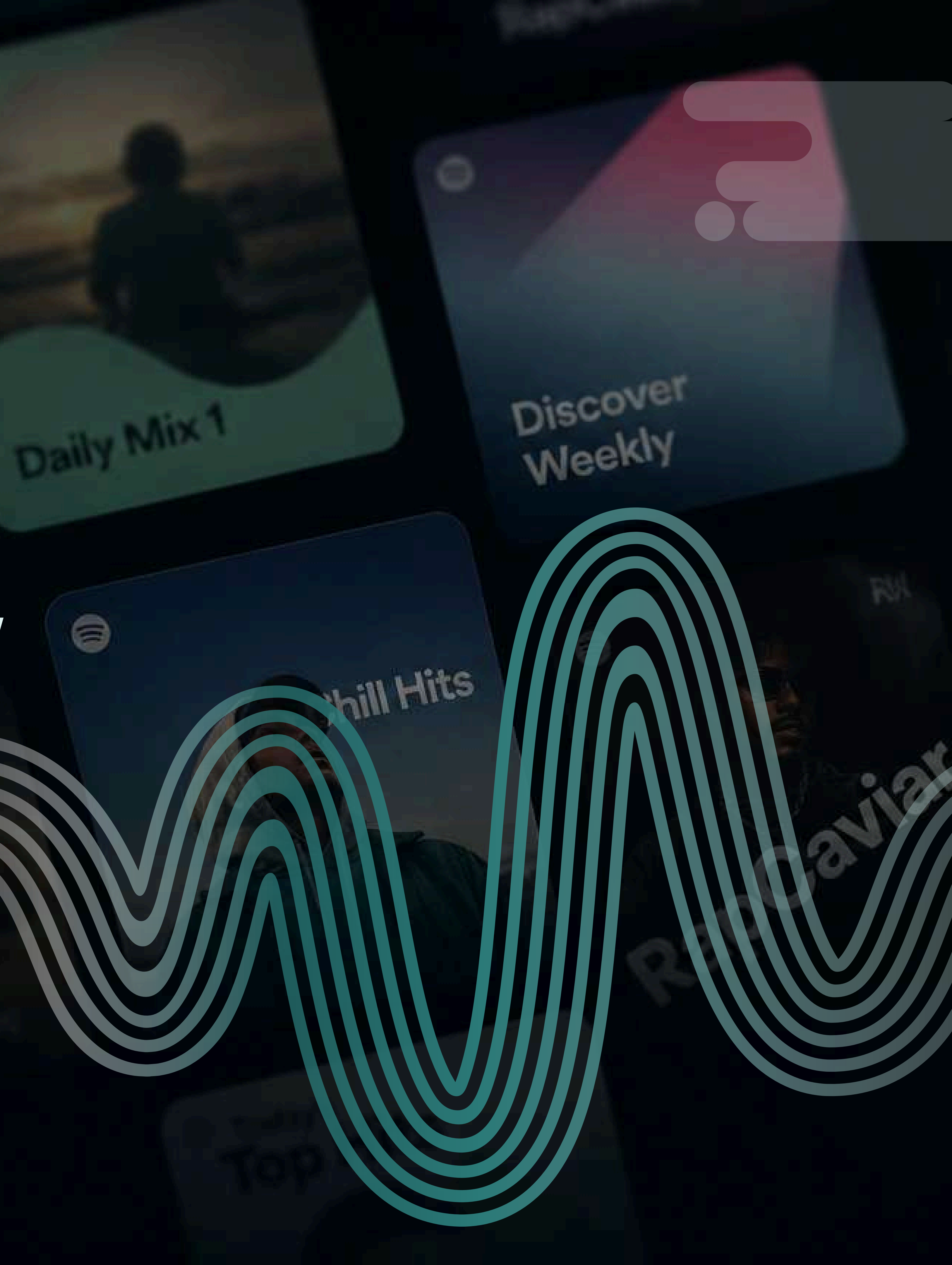
Designing Intent-Based Music Discovery

to Reduce Choice Overload

Helping users find the right music
instead of just more music.

Anunay Pal



Executive Summary

Spotify has transformed how people consume music through personalized recommendations. Yet despite having access to over **100 million songs**, many users still spend several minutes deciding what to listen to before they actually start listening.

> The challenge isn't content availability.
> The challenge is helping users discover music that matches their **current intent**.

Instead of asking,
"What songs might this user like?"
this case study explores a different product question:

> **How might Spotify understand why a user opened the app and recommend music that instantly matches their context, mood, and activity?**

Through user research, behavioral analysis, and competitive benchmarking, I designed an **Intent-Based Discovery Layer** that enables Spotify to recommend music around real-world situations rather than only listening history.

Deliverables

- User Research
- JTBD Analysis
- Competitive Benchmark
- Product Solution
- Wireframes
- Success Metrics
- PRD

Why **Spotify?**

Spotify isn't just a music streaming platform.

t's a daily companion used while studying, working, exercising, travelling, relaxing, celebrating, and even sleeping.

Yet despite its sophisticated recommendation engine, users often spend considerable time searching before listening.

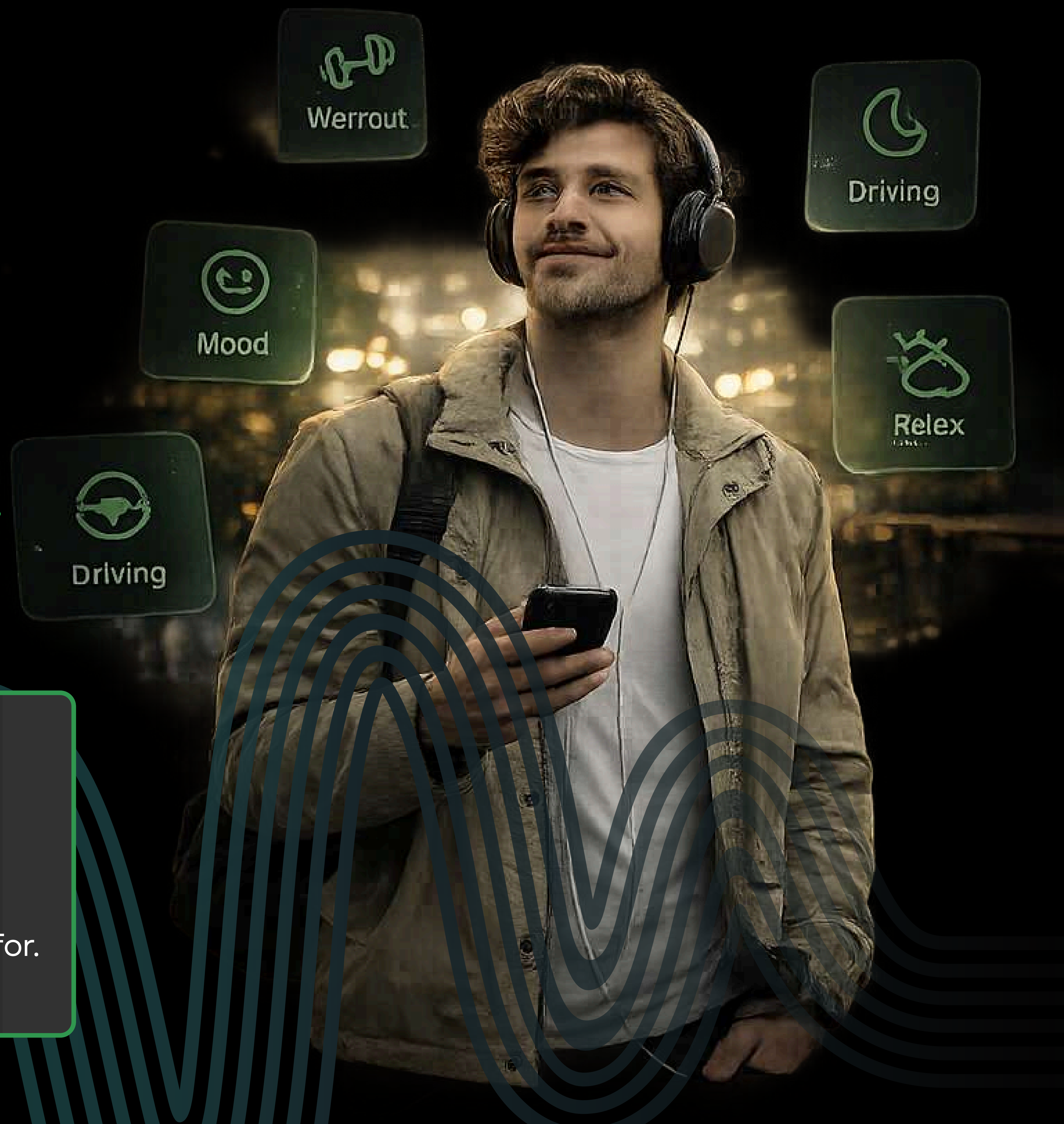
The product already understands what users have listened to.
But can it better understand why they opened Spotify today?

That question became the foundation of this case study.

Why this problem matters

More than **100 million** songs create decision fatigue.

- Users often skip multiple tracks before settling on one.
- Recommendations become repetitive over time.
- Search requires users to know exactly what they're looking for.
- Intent is often stronger than listening history.



Problem Statement

Spotify excels at recommending songs similar to what users already enjoy. However, users frequently open Spotify with a specific goal rather than a specific song.

Users open Spotify to:

- Focus while studying
- Boost productivity
- Get motivated at the gym
- Enjoy while driving
- Relax on a rainy evening
- Heal after a breakup

Current discovery experience falls short:

Waste time before listening
Users spend several minutes searching or browsing.

Recommendations feel repetitive
Over-exposure to similar songs and artists.

Search requires too much effort
Users must know what they want in advance.

Algorithms prioritize familiarity
Less room for discovery and serendipity.

Doesn't adapt to real-time context
Mood, activity, time and environment aren't fully considered.

Problem Statement

Spotify helps users find songs they already like, but struggles to quickly understand the **context** behind why users opened the app.

As a result, users spend more time searching, experience choice overload and miss out on music that could perfectly match their intent in the moment.

The impact:

30%
of users skip multiple songs before settling on one.

27%
of users feel Spotify recommends the same type of music.

41%
of users agree that discovering new music is getting harder.

18%
drop-off sessions within the first 2 minutes.

Spotify at Scale

Understanding the Product before solving the problem

Spotify isn’t just a music streaming platform.
It’s a daily companion that powers millions of moments-whether users are working, studying, exercising, commuting, relaxing or sleeping.

Today, Spotify delivers highly personalized recommendations using listening history, collaborative filtering anfd machine learning.

Yet one challenge continues to exist:
Users often know how they feel, but not what they want to play.

As Spotify’s library continues to grow, discovering the right content has become more difficult than accessing content itself.

This creates an oppurtunity to shift recommendations from being history-driven to becoming **intent-driven.**

Key Product Facts:

- **100M+** songs
- **6M+** podcasts
- **675M+** Monthly Active Users
- **268M+** Premium Subscribers
- **180+** Markets
- One of the world’s largest recommendation systems

> The problem isn’t content availability.
> The problem is helping users find the right content instantly.

Spotify

User **Research**

Understanding Real Listening Behavior

Instead of assuming why users struggle, I analyzed how people actually discover music on Spotify across different situations.

The goal wasn't to understand what users listen to, but **why they open Spotify** in the first place.

1. User Interviews

18 in-depth interviews

- Students
- Working Professionals
- Gym-Goers
- Music Enthusiasts
- Casual Listeners

Goal: Understand intent, context, emotions and discovery challenges.

2. App Store Review

250+ reviews analyzed

Top recurring themes:

- Discovery
- Recommendations
- Search
- Daily Mix
- Smart Shuffle

Goal: Identify large-scale pain points and patterns.

3. Behavioral Observation

15+ observation sessions

Observed how users searched for music before selecting something to play.

Focused on:

- Search behaviour
- Playlist browsing
- Recommendation usage
- Time-to-first-play

Goal: Map actual behaviour vs. what users say.

4. Competitive Benchmarking

4 major music platforms

Compared Spotify with:

- YouTube Music
- Apple Music
- Amazon Music
- Tidal

Goal: Understand gaps and oppurtunities in the market.

Key Research Question

“What job is the user hiring Spotify to do at this exact moment?”

Outcome ->

Four clear user **personas** emerged.

User Persona

Four different intent patterns

Student

"I need music that helps me study."

Pain Points

- Distracting recommendations
- Too many choices
- Time wasted searching

Needs

- Focus music instantly

Working Professional

"I've had a stressful day."

Pain Points

- Doesn't know what mood matches
- Wants somwthing calming

Needs

- Emotion-aware recommendations

Gym User

"I need energy"

Pain Points

- Existing playlists become repetitive

Needs

- Dynamic-workout playlists

Causal Listeners

"I just want something good."

Pain Points

- Doesn't know what to search
- Overwhelmed by options

Needs

- Intent-first discovery

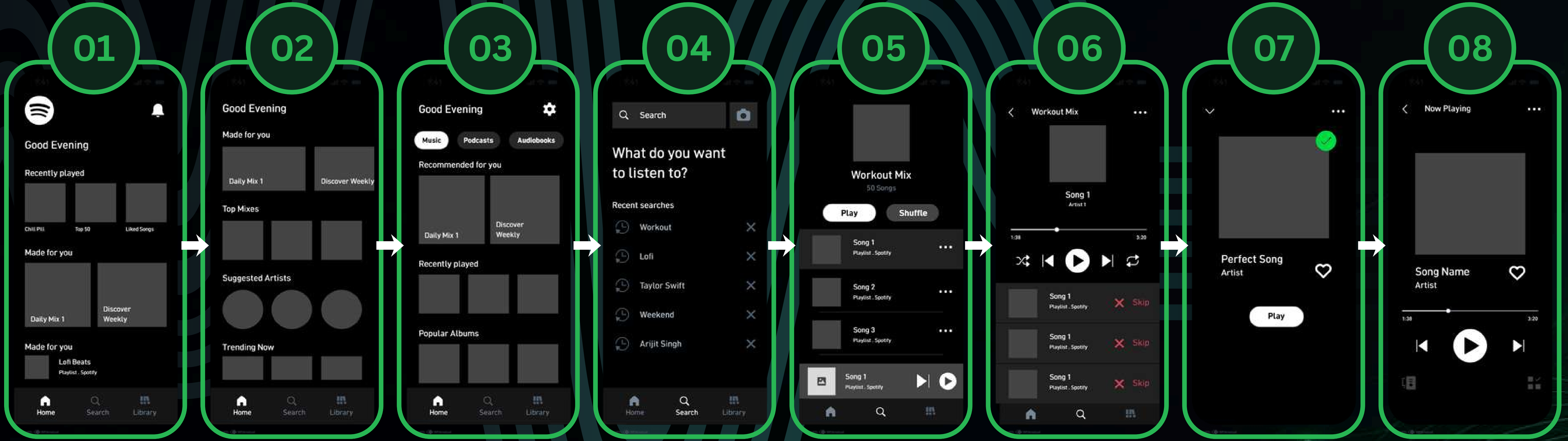
Key Insight

Different users open Spotify for **completely different reasons.**
The current recommendation engine treats all discovery sessions **almost the same.**

Spotify

Current User Journey

Where Friction Begins



01	02	03	04	05	06	07	08
Opens Spotify	Looks at Home Screen	Scrolls Recommendations	Searches Again	Opens Playlist	Skips Multiple Songs	Finally Finds One	Starts Listening
	Too many choices	Still searching	User switches from recommendations to search		Still not the right mood	Finally found something	Listening begins

Friction Points

Choice overload
Too many options on the home screen.

Search dependency
Users often search because recommendations don't fit.

Generic recommendations
Lack of context leads to irrelevant suggestions.

Mood mismatch
Recommendations don't match current mood/activity.

Repetitive playlist
Playlists feel stale and predictable.

Key Insight

The biggest friction doesn't happen while listening. It happens **before users play their first song.**

Spotify optimizes music consumption, but **not music selection.**

Spotify

Root Cause Analysis

Why Spotify still creates decision fatigue.

Decision Fatigue

Users spend too much time deciding what to play before they actuslly start listening

1 Recommendation Model

- Prioritize listening history.
- Slow to adapt to changing intent.
- Similar artists dominate suggestions.

2 Home Screen Experience

- Static recommendations.
- Same playlists repeatedly surface.
- Limited contextual personalization.

3 User Context Missing

Spotify doesn't understand:

- Activity
- Time of day
- Mood
- Intent
- Location

4 Discovery Experience

Users must search manually

↓

High effort

↓

Less exploration

↓

Lower satisfaction

Root Cause

Spotify **predicts taste**, but **not current intent.**

Spotify

JTBD + Key Insights

Understanding the jobs users are trying to get done when they open Spotify.

Jobs To Be Done (JTBD)

When I'm studying
Help me start a distraction-free session **within seconds.**

When I'm working out
Help me find **high-energy** music without searching.

When I'm driving
Help me play safe, hands-free music **instantly**.

When I'm relaxing
Help me discover calming music that **matches my mood.**

Key Insights

Insights 1
Users open Spotify with an intention, not with a specific song.

Insights 2
Mood changes faster than listening history.

Insights 3
Users want less decision-making, not more recommendations.

Insights 4
Discovery works best when recommendations match the current context.

Core Oppurtunity

Instead of studying
"What songs might this user like?"

Spotify should ask
"Why did this user open the app right now?"

Spotify

Competitive Benchmarking

Most platforms personalize your music. Now understand your intent.

Feature	Spotify	Apple Music	Youtube Music	Amazon Music
Personalized Mixes (Discover Weekly, Daily Mixes)	✔	✔	✔	✔
Mood Based Playlists	✔	✔	✔	✔
All Recommendations	✔	✔	✔	Limited
Activity / Context Awareness	Partial	Limited	Limited	Limited
Intent Detection before recommending	✘	✘	✘	✘
One-Tao Context Selection	✘	✘	✘	✘
Hands-free / Quick Srart Experience	Partial	Partial	Partial	Limited

The Oppurtunity:

No major streaming platform explicitly asks or predicts a user's immediate intent before recommending music. Spotify can lead with an intent-Based Discovery Layer.

Spotify

Product Solution: Spotify Intent

A lightweight layer that understands why you opened Spotify and delivers music that fits the moment.

- **Intent Cards**
  Tap an intent that matches your goal, mood or activity.
- **Smart Context Detection**
  Uses time, location, motion, connected devices and listening signals to detect context in the background.
- **Instant Playlist Generation**
  AI curates the most relevant playlist for your intent in less than a second.
- **Continuious Learning**
  Every interaction helps Spotify learn your preference for each intent and moment.

Faster music selection

Lower search effort

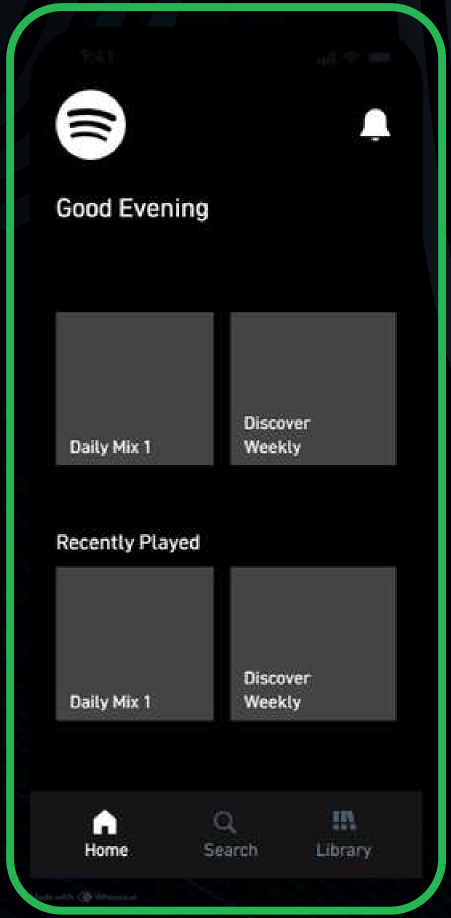
Better Discovery

Higher Engagement

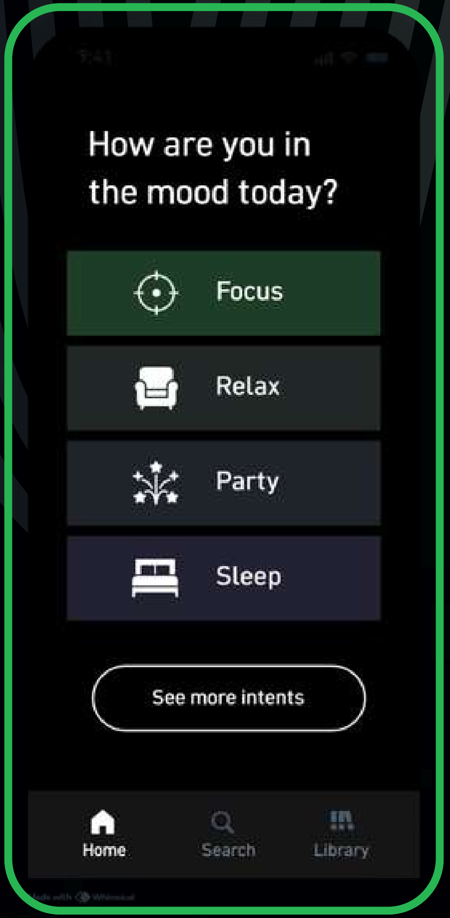
Spotify

Future User Flow & Wireframes

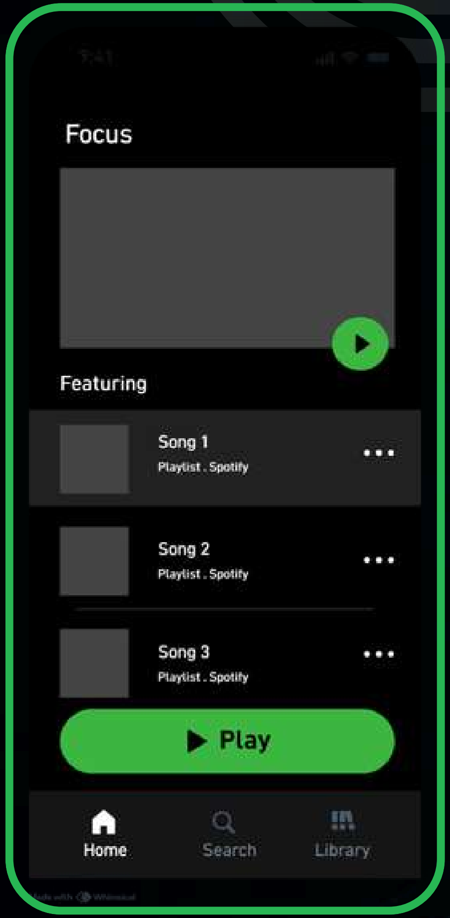
Wireframes (Key Screens)



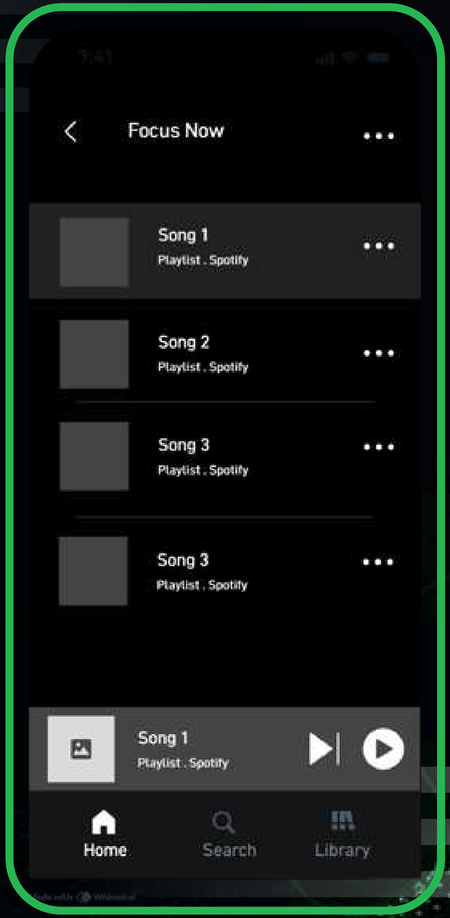
Home Screen



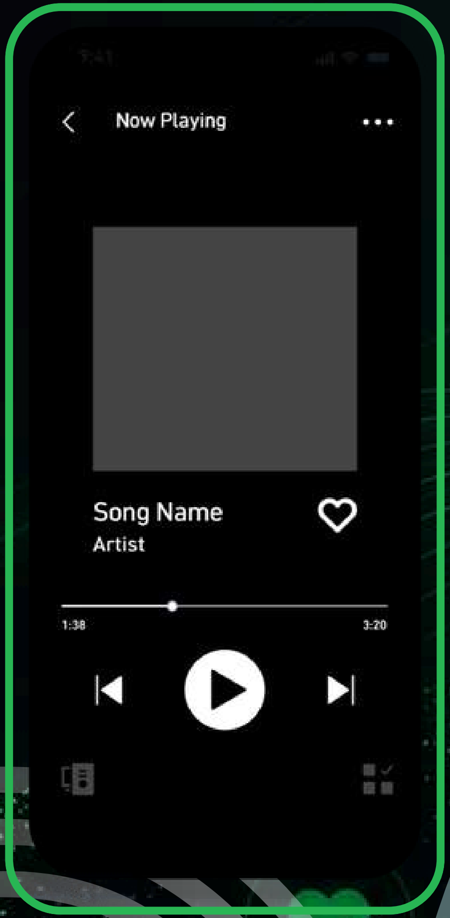
Intent Selection



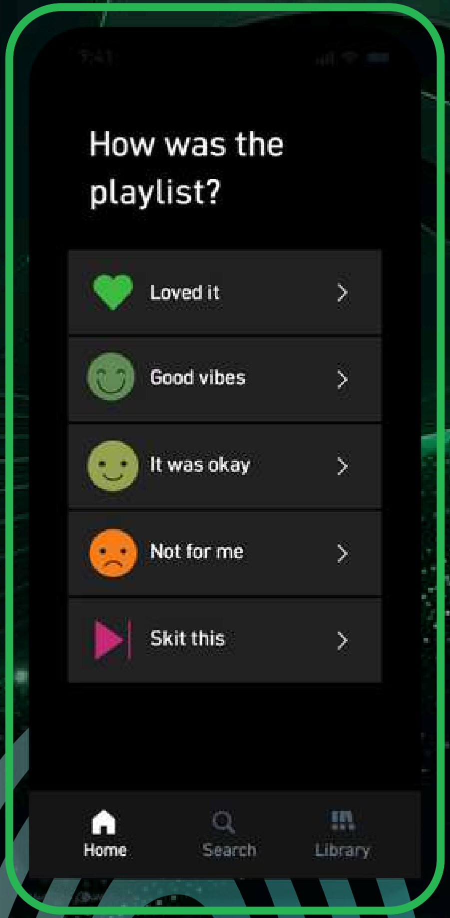
Playlist Details



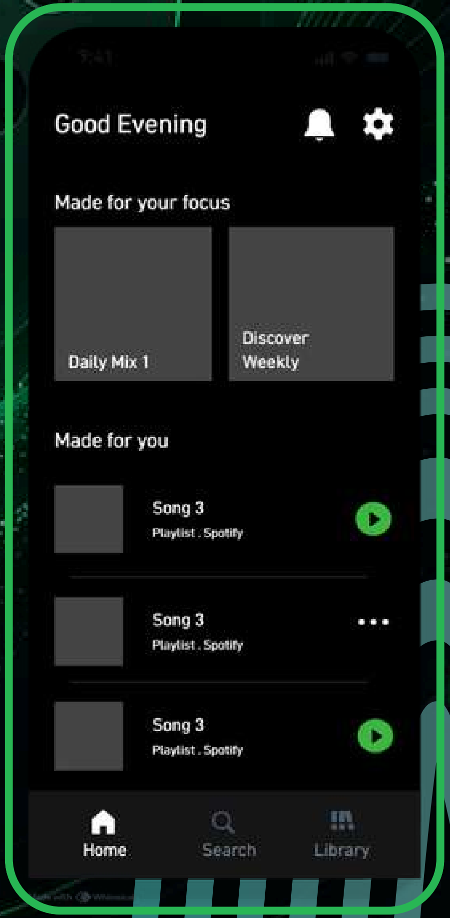
Playlist



Skips Multiple Songs



Feedback Prompt



Starts Listening